Human Compression: current audit protocol

Human Compression project

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1 Corpus and current scope

The active corpus consists of 106 accessible originals from a fixed historical catalogue of 161 titles. New-title intake and retrieval of the 55 archived unavailable originals are outside the current programme. An accessible original remains a mathematical target while its recovery is unfinished. A complete authored rendering can supply a usable source without a publisher PDF, provided its edition and representation are identified explicitly.

The current partition is

$$106 = 52 \text{ measured recoveries} + 53 \text{ mathematical targets} + 1 \text{ editorial}.$$

L022 is the editorial: complete reading found no principal mathematical result to recover. It remains an accessible original, with no measurement factor or mathematical recovery credit. The other 55 historical originals remain archived. The register counts originals, not theorem nodes, contracts, editions or files.

The accepted overview v18 reports nineteen complete-topic factors and fifty-two conditional application factors. Its graphs show the whole common theory and all applications. The exact current records are the corpus register, the overview pointer and the manuscript catalogue. The six-page overview gives the values; this protocol adds no new measurement or scientific admission.

2 The mathematical object being recovered

Before comparing lengths, each original receives an edition-bound source contract. Its principal results, parameter ranges, exceptional cases, geometric identifications and computational obligations are stated explicitly. A complete application must prove the relevant hypotheses of the common theory and supply all local arguments needed for those outputs. Retained classical theorems, classification tables and catalogue constructions remain named premises.

A false source assertion is recorded as a correction, not as a theorem proved with its original wording. Unrecovered ancillary statements, illustrations, classifications and open questions remain explicit. Recovery of the principal contract does not assert reproduction of every assertion in the source. Consequently, a full-source length comparison is a comparison for the declared principal outputs, not a claim that every source passage has an equivalent replacement passage. The source still contributes its complete counted manuscript, including ancillary exposition and references.

Scientific acceptance, input acceptance and numerical acceptance are separate. An author packet, distinct review and later admission retain their exact bytes and roles. Earlier candidate labels in a frozen packet are interpreted through its subsequent admission. Existing proofs can be reorganized or reused, but moving a proof between files does not itself establish a new mathematical result.

3 Whole topics and conditional applications

For a topic T, write $L_1, \ldots, L_k$ for the complete source manuscripts, U_T for its entire common theory, and $A_1, \ldots, A_k$ for its complete applications. An application includes every required written companion. Under the topic's fixed manuscript counting convention c_T, the whole factor is

$$R_T = \frac{\sum_{i=1}^{k} c_T(L_i)}{c_T(U_T) + \sum_{i=1}^{k} c_T(A_i)}.$$

All common papers and additions actually used in the topic belong to U_T. The whole common theory is charged once within that topic. Separate topics pay their own common theories in full, even when their arguments overlap. There is no cross-topic pooled factor or unrecorded discount for previously written work.

The individual comparison is

$$r_{T,i} \mid \mathcal{K}_T = \frac{c_T(L_i)}{c_T(A_i)}.$$

Here $\mathcal{K}_T$ includes the *whole* common theory, the declared required programs and data, and any retained mathematical premises. Their acquisition costs and unknown costs are reported separately. Any additional sibling-application premise must be included explicitly in the acquisition boundary; it cannot be supplied for free. The accepted current A edition, for example, pays all 23 common pages and all nine applications in its whole-topic comparison, rather than allocating only selected common paragraphs to each one.

A factor greater than one records shortening in the stated coordinate; a factor below one records expansion. A conditional saving may coexist with a whole-topic expansion. Such results are retained without changing the metric or dropping applications to obtain a preferred value. These factors are not measures of historical effort, mathematical originality or minimum proof complexity.

4 Text representations and sensitivity

The input protocol is fixed and reviewed before factors are evaluated. It binds the exact source edition, every complete replacement PDF, document boundaries, software masks, font evidence, normalization and units. Titles, abstracts, body text, captions and references remain included. Exact running furniture can be removed only in the declared selected-text coordinates; an untrimmed coordinate preserves it. Some accepted topics use that untrimmed coordinate as primary. The overview retains each topic's accepted choice rather than silently replacing them with a single new convention.

The usual sensitivity family combines raw untrimmed, raw selected and calibrated selected text with NFC and NFKC normalization and five units: whitespace-separated words, alphanumeric runs, non-whitespace characters, characters with collapsed whitespace, and UTF-8 bytes with collapsed whitespace. The exact definitions and any additional representation sensitivities are those of each frozen input protocol. A display-rounding choice changes no exact fraction. A near-unit factor may reverse under another accepted unit; this is reported rather than suppressed.

Calibration identifies the actual PDF font dictionary and encoding, the text event and the visible glyph. Font names alone need not identify an encoding. Repairs, including visible glyphs extracted as whitespace, require occurrence and outline evidence. A chosen proxy is declared as a proxy. Raw extraction remains available for sensitivity; calibration is not a transcription of all visual mathematical content.

Complete authored renderings, including the accepted N001 and L108 sources, are distinguished from unavailable original PDFs. Website navigation or duplicated reference widgets are not article text. Images, diagrams, layout and unextracted labels have unknown cost; their

absence from text extraction is not a zero-cost estimate. In particular, the L035 source contains scientific image content not represented by its extracted text. All current factors are subject to their documented representation limits.

5 Symmetric software treatment and proof artifacts

The current scalar is manuscript-only. Concrete executable program and implementation-listing costs are excluded symmetrically from source and replacement. This rule applies to required computations as well as optional validation. Written mathematical algorithms and proof prose remain counted on both sides. A software role alone cannot justify counting a source listing while omitting the corresponding replacement program.

For L008, L028 and N006, independently reviewed masks remove the exact embedded implementation blocks, retaining surrounding mathematical statements and exposition. Their restored manuscript comparisons use those accepted boundaries. The listing audit and matched input packet preserve the locators and sensitivity evidence. Earlier one-sided software-inclusive values remain historical observations and are not current comparisons.

Required computations remain proof premises. Each topic inventories every whole required program, certificate, mathematical data file and decoder, with separate roles for optional validation, execution drivers and partial formal artifacts. Reused identical content may be counted once within an actual topic dependency closure, with every occurrence and role disclosed. No file is pruned merely because a shorter word certificate or a previously run search is available. Retained external representations and catalogues remain explicit trust and acquisition boundaries.

A printed program appendix and its raw files are alternative representations, not two additive copies of one program. A printed appendix need not contain a required data file; the data still belongs in the complete inventory. Historical source software and full external proof costs are generally unknown. Accordingly, there is no current software-inclusive factor. No missing cost is interpreted as zero, and incompatible units such as pages and bytes are never added.

6 Proof dependencies and formal evidence

A logical graph records actual prerequisite-to-use relations with source and proof locators. Mathematical premises, definitions, retained theorems, computations, validation, attribution and operational dependencies have distinct roles. A citation's presence does not by itself prove logical dependence. A graph accepted for an earlier edition is not automatically a complete graph for an enlarged topic.

Where passage costs have been accepted, each counted passage has one owner. For an output t, let $D(t)$ be its prerequisite closure and $P(v)$ the passage owners used by node v. The known passage cost is

$$W_{\text{known}}(t) = \sum_{p \in \bigcup_{v \in D(t)} P(v),\ c(p)\ \text{known}} c(p).$$

The union prevents repeated charging of shared passages. Unknown costs remain listed separately; context outside that closure is still paid in the whole-manuscript comparison. Parent and child views must preserve this total. Raw node counts are not compression factors.

For an explicit acquisition order and a previously acquired library K_{i-1}, a sequential account can report the new closure cost $c(D_i \setminus K_{i-1})$. It requires actual accepted dependencies and costs; it does not establish a sawtooth pattern in advance or alter separate topics' whole-common charges.

Lean evidence is limited to the actual compiled declarations and their stated hypotheses, dependencies and axiom reports. A theorem assuming an unproved recognition or geometric

hypothesis does not formalize that hypothesis. Successful compilation and declaration counts do not establish complete source formalization. Finite certificates likewise require an explicit argument from the checked computation to the claimed result. The current numerical comparison adds no complete-Lean or conjecture-answer claim.

7 Reproduction and historical records

Current TeX/PDFs, source maps and admissions are routed through the public catalogue. Original source assets are separately supplied where redistribution is unavailable. Portable input and arithmetic checkers record their exact versions, hashes and output boundaries. The canonical builder uses each manuscript's declared compiler and full TeX inputs. Repository integrity checks, formal compilations and expensive finite replays are different operations; evidence states which actually ran.

Historical editions remain immutable. The preceding README and preceding protocol retain the old chronology and comparisons. The intake record, archived notes and earlier overview admissions remain evidence for their stated editions. A documentation update supersedes reader-facing status and policy statements without rewriting those records. Separately reviewed question answers retain their own exact scope; an absent historical proof has no measured length and does not produce an infinite compression factor.